



Defence Innovation Portal (IDEaS and NATO DIANA)

Component 1a

Proposal

A proposal describes a proposed solution aimed at establishing the knowledge and technological advantage needed to address Canada's defence and security interests for the selected S&T Challenge.

The responses provided by the Offeror regarding the submission of the Technical and the Financial Proposal will form the Offeror proposal in its entirety.

Note: The following acronyms are used throughout the proposal and are defined as:

SC = Screening Criteria; **MC** = Mandatory Criteria; **PRC** = Point-Rated Criteria

Company name: zax analytics

Company representative name: Steven Ness

Proposal Reference Number: CP6-132484

Category of Submission: Replacement Submission

Proposal reference number being replaced:: CP6-132296

Proposal Submitted Date: 2026-06-01 09:03 PM

Proposal Title
Planetar: A Learned Cross-Modal AI Model for Maritime Dark-Vessel Detection

1 - Science and Technology (S&T) Challenge

A. S&T Challenge of the Proposed Solution

Identify the S&T Challenge being addressed by this proposal.

S&T Challenge of the Proposed Solution
Multi-modal AI for advanced situational decisions

2 - Project Description

A. Project Synopsis

Provide a brief synopsis of the proposed solution.

Note: A one-page PDF or JPEG file can be uploaded to support your synopsis. (Max. 5MB)

Project Synopsis

Dark vessels — vessels that disable their AIS transponders to evade tracking — are a central capability gap for maritime situational awareness. Illegal fishing, sanctions evasion, ship-to-ship transfers, and Arctic sovereignty incursions all depend on AIS going dark. Today this is handled with per-modality pipelines and explainability bolted on afterward. Planetar is a learned cross-modal AI model that closes this gap. It encodes six streams — AIS, synthetic-aperture radar (SAR), electro-optical imagery (EO), hydrophone audio, radio-frequency emissions, and textual maritime reports — into a shared representation where one vessel's observations resolve to a single identity, even after its AIS beacon goes dark. The model trains itself: while a vessel broadcasts AIS, that identity labels the concurrent radar, optical, acoustic, RF, and text observations, so the model learns the cross-modal association and can re-identify the vessel once it goes dark. Every result carries a calibrated confidence and an evidence trail to the raw inputs, with a human analyst in the loop. At project start the supporting platform is working open-source code: a real-time, provenance-tracked data substrate, an entity graph (a production successor to an 18-month prototype), an analyst console, and four per-modality detectors (AIS, SAR, EO, acoustic). The new research for Component 1a is the learned fusion model itself, its uncertainty calibration, and the text and RF encoders. End-state: a synthetic Salish Sea scenario — a vessel disables AIS during a Sentinel-1 satellite pass, is re-detected by SAR, EO and hydrophone, and is re-identified by the model — fully replayable and live at planetar.ca, with public-benchmark evaluation. Solo Zax Analytics execution; public data only; no government-furnished property, no field deployments, no classified content.

B. Keywords

Provide 3 to 5 keywords related to your project to help us index the solutions and assign the appropriate technical and scientific evaluators to the proposal.

Keyword
multi-modal sensor fusion
maritime dark-vessel detection
self-supervised cross-modal learning
uncertainty quantification and calibration
entity resolution and provenance

C. Project Overview

Provide a project overview, which is a synopsis of project's S/T Merit, Novelty and Innovation, Impact, Feasibility and Approach.

Planetar is a learned cross-modal AI model for maritime situational awareness. It encodes heterogeneous sensor and text streams into a shared representation where one vessel's observations resolve to a single, calibrated, explainable identity — even when the vessel disables its Automatic Identification System (AIS) to go dark. The Component 1a flagship is dark-vessel detection; the supporting real-time, provenance-tracked data platform is already built and open-source. Scientific and Technical Merit (PRC-1): The core method is a self-supervised cross-modal fusion model. While a vessel broadcasts AIS, its identity labels the concurrent synthetic-aperture radar (SAR), electro-optical (EO), acoustic, radio-frequency, and text observations; the model learns that association and applies it to re-identify vessels once AIS is dark — dissolving the scarce-label barrier that blocks supervised maritime fusion. Per-modality uncertainty is propagated and calibrated with conformal prediction for distribution-free confidence bounds. The applicant's peer-reviewed semi-supervised deep learning on unlabelled acoustic archives is direct evidence the approach is achievable. Novelty and Innovation (PRC-2): Self-supervised, learned cross-modal dark-vessel re-identification across this modality set is, to the applicant's knowledge, absent from the literature. Fusion is learned and adaptive — not rule-based aggregation — with attention-grounded explanations and a human analyst in the loop. Impact (PRC-3): Closes the operationally costly AIS-off capability gap (illegal fishing, sanctions evasion, Arctic sovereignty incursions). Restoring track continuity on vessels that evade monitoring reduces a surveillance vulnerability and speeds decision-making. Open-source on commodity hardware it is addressable for Canadian Armed Forces (CAF) interoperability with allies. Feasibility and Approach (PRC-4): The work runs over two milestone stages on public data. Milestone 1 hardens the data substrate and the six-modality ingest (adding text and RF), builds the per-modality encoders and the AIS co-occurrence training set, and trains the self-supervised cross-modal fusion model. Milestone 2 calibrates and evaluates uncertainty on AIS-off re-identification, builds the explainable human-in-the-loop analyst surface, and delivers a replayable Salish Sea demonstration plus a live system at planetar.ca the evaluator can operate, with public-benchmark evaluation. Solo Zax Analytics execution. No government-furnished property, no field deployments, no classified content.

MC-1 - Technology Readiness Level (TRL) of the Proposed Solution

Select and describe the TRL of the proposed solution before starting the project and the projected end-state under the contract.

Note: [Information on Technology Readiness Levels \(TRLs\)](#).

TRL of the solution before starting the project: **TRL-2: Technology concept and/or application formulated**

Describe the research and development activities that have taken place to bring the proposed solution to the stated TRL.

Current TRL: 2. The solution is a learned, self-supervised cross-modal fusion model that re-identifies a vessel after it disables its AIS transponder. The concept is fully formulated — a concrete architecture with substantial component-level evidence from the research and development below — but the integrated model is not yet trained or demonstrated; building and demonstrating it is the work the project funds, advancing the solution to TRL 3. Per-modality ingest and detection are already built and open-source (about 6,000 lines, broker-integrated): a live AIS service, a Sentinel-1 synthetic-aperture-radar detector validated on a 433-megapixel scene, an electro-optical detector, and a passive-hydrophone detector. These are the encoders the fusion model consumes, so the perception front-end already exists rather than being assumed. Entity resolution with provenance is likewise built: a zero-dependency production service, with 30 automated tests passing, resolves cross-modal observations to canonical vessel identities while preserving full lineage. It builds on an eighteen-month reference implementation and is a substantial development beyond the integrated-entity-view architecture of US Patent 10,936,582, on which the applicant is a named inventor (Salesforce-assigned), not the patent holder. The deep-learning approach the model rests on is evidenced by the applicant's peer-reviewed work in the same regime it will operate in — scarce labels over continuous, unlabelled streams: semi-supervised deep learning at archive scale on hydrophone data (ORCA-SLANG, Interspeech 2021; and Sattar et al., IEEE PacRim 2011, evaluated on Ocean Networks Canada data at 95% accuracy). The planned self-supervised, joint-embedding design follows the established I-JEPA family of methods. Underpinning these, a real-time, provenance-tracked substrate — a message bus, a zero-copy typed envelope, and a multi-pane analyst console — already runs as a live system and gives every inference an inspectable causal evidence chain with bit-exact replay. Together, this work formulates the concept and de-risks it: the architecture, the encoders, the supervision signal, and the deployment surface all exist. The six months is therefore spent on the novel model itself rather than on plumbing — building, training, and calibrating it, and demonstrating it end-to-end on public data and as a live system at planetar.ca — advancing the solution from TRL 2 to TRL 3.

Select the projected end-state TRL of the proposed solution under the contract.: **TRL-3: Analytical and experimental critical function and/or characteristic proof of concept**

Provide a high-level explanation with relevant examples to justify your anticipated TRL at the end-state by applying the TRL definitions provided above.

At project start the solution — a learned, self-supervised cross-modal fusion model — is at TRL 2: the concept is formulated with strong component evidence (four working per-modality detectors, an AIS co-occurrence supervision signal, a built entity-resolution graph, and the applicant's peer-reviewed semi-supervised deep learning), but the integrated model is not yet built or demonstrated. The anticipated end-state is TRL 3 — analytical and experimental critical function and/or characteristic proof of concept — reached when the model is built, trained on public data, calibrated, and demonstrated end-to-end. The demonstration is twofold: a controlled, reproducible synthetic Salish Sea scenario in which a vessel disables AIS during a Sentinel-1 satellite pass, is re-detected by SAR, EO and hydrophone, and is re-identified by the model with a calibrated confidence and a full evidence chain, replayable bit-for-bit from the write-ahead log; and a live system at planetar.ca that evaluators can drive themselves during evaluation, exercising the same model on the same public data. A running system an evaluator can operate is the defining characteristic of a demonstrated proof of concept. The work proceeds in two milestone stages. Milestone 1 hardens the data substrate, stands up text and RF ingest, builds the per-modality encoders and the AIS co-occurrence training set, and trains the self-supervised cross-modal fusion model. Milestone 2 propagates and conformally calibrates uncertainty and evaluates AIS-off re-identification, ships the explainable human-in-the-loop analyst surface, and delivers the integrated Salish Sea demonstration with the live planetar.ca deployment and a public-benchmark evaluation. Each detection traces to its inputs, each re-identification carries a calibrated confidence interval, and the full chain can be independently re-run from the open-source repositories — analytical and experimental evidence, not assertion. This is a clean one-level advance from TRL 2 to TRL 3, squarely inside the Component 1a Conceive bracket (TRL 1–3). TRL 4 — component or breadboard validation in a laboratory environment — is the natural Component 1b objective this proof of concept sets up.

MC-2 - Alignment of Proposed Solution to S&T Challenge

Describe the proposed solution and how it addresses and aligns to the S&T Challenge identified above by including a description of the scientific and technological basis upon which the solution is proposed, and clearly justify how the solution meets the Essential Outcomes.

MC-2 - Alignment of Proposed Solution to S&T Challenge

Planetar is a learned cross-modal fusion model for maritime domain awareness. Six heterogeneous streams — AIS (kinematic + text fields), Sentinel-1 SAR, electro-optical, passive acoustic, RF emissions, and textual maritime reports (notices-to-mariners / port-state / OSINT) — are encoded into a shared spatiotemporal embedding in which observations of the same vessel resolve to one calibrated, uncertainty-scored identity, even after its AIS beacon goes dark. The model is trained self-supervised: during AIS-on periods, a broadcasting vessel's identity labels the concurrent SAR/EO/acoustic/RF/text observations in its space-time neighborhood; the model learns that cross-modal association and applies it to re-identify AIS-off ("dark") vessels — the Challenge's Maritime Task Group Operations example. Every output carries calibrated confidence and an attention-weighted evidence chain, with a human analyst in the loop. This serves the CAF Digital Campaign Plan and DND/CAF AI Strategy goals of learned, explainable multi-domain fusion for ISR and C2.

Scientific and technological basis. (i) Deep multimodal representation learning and learned association — per-modality encoders projected to a shared metric space, a transformer fusion head over concurrent observations, trained self-/semi-supervised on AIS co-occurrence. The applicant's peer-reviewed semi-supervised deep learning on unlabeled acoustic archives (ORCA-SLANG, Interspeech 2021; Sattar et al. PacRim 2011, 95 % on Ocean Networks Canada data) is direct evidence the approach is achievable. (ii) Uncertainty propagation — per-modality evidential heads carried through fusion, then conformal calibration for distribution-free coverage. (iii) Entity resolution + dynamic knowledge graph for persistent cross-domain tracking. (iv) Native explainability and provenance — each inference renders its causal evidence for operator trust and accreditation. A working real-time, provenance-tracked substrate the applicant has already built makes the model deployable at the edge (SWaP) and bit-exact replayable. Essential Outcome compliance. The Essential Outcome requires an AI model fusing 2 heterogeneous types (e.g. sensor, text, RF) into classifications / detections / correlations. Planetar's model fuses six types spanning exactly sensor, text, and RF, producing: detection (vessel present), classification (vessel class), and correlation (cross-modal identity / dark-vessel re-ID) with calibrated confidence and per-modality evidence. Fusion is learned and adaptive — a trained cross-modal association model, not rule-based aggregation — and adapts its perception front-end to SWaP and threat conditions at the edge. The two-type bar is cleared with margin, against the Challenge's own sensor / text / RF example.

PRC-1 - Scientific and/or Technical (S/T) Merit of Proposed Solution

Articulate how your proposed solution meets the Scientific and/or Technical Merit criterion.

PRC-1 - Scientific and/or Technical (S/T) Merit of Proposed Solution

(a) Sound S/T evidence. The approach rests on peer-reviewed deep-learning method, working implementations, and reproducible engineering. Learned multimodal fusion: shared-embedding cross-modal representation, attention over concurrent-observation sets, metric-learning re-identification, and joint-embedding self-supervision are established methods, which the 1a composes for maritime dark-vessel re-ID. Semi-supervised deep learning on continuous unlabelled streams: the applicant co-authored ORCA-SLANG (Interspeech 2021) and Sattar et al. (PacRim 2011, on Ocean Networks Canada data) — the scarce-label, continuous-stream regime the maritime model works in. Calibrated uncertainty: per-modality evidential heads with conformal prediction for distribution-free coverage. Entity resolution with provenance: the applicant's ~18-month entity-resolution implementation maps cross-modal observation to identity with lineage. Dark-vessel tractability: public DoD-component precedent (DIU xView3; Park et al., Science Advances 2020) establishes AI advantage over classical baselines. Auditory representation learning: Ness, Walters, Lyon (2012) (Walters/Lyon at Google Research), relevant to the acoustic encoder. Real-time / edge engineering: a provenance-tracked substrate runs the model on commodity hardware (SWaP) with measured, reproducible performance. (b) State-of-the-art. The model advances current practice on the axes the Challenge names. Learned, adaptive fusion replaces rule-based aggregation, and adapts at the edge via a runtime-rewritable perception graph. Self-supervision from AIS co-occurrence removes the scarce-label barrier that blocks supervised maritime fusion — to the applicant's knowledge new in this domain. Uncertainty and explainability are system properties: propagated evidential uncertainty, conformal coverage, and an attention-grounded causal evidence chain accreditable for operator trust, rather than post-hoc XAI. Acoustic methodology: the applicant's archive-scale semi-supervised hydrophone ML extends from cetacean-call to vessel-signature discrimination, a structurally identical learning problem.

PRC-2 - Novel & Innovative Solution

Articulate how your proposed solution meets the Novel and Innovative criterion.

PRC-2 - Novel & Innovative Solution

(a) New knowledge / technology. (a.1) Self-supervised cross-modal re-identification — the core contribution. A learned model uses AIS-on co-occurrence as a free training signal: when a vessel broadcasts AIS, its identity labels the concurrent SAR/EO/acoustic/RF/text observations in its space-time neighborhood, letting the model learn the cross-modal association; at inference it re-identifies vessels whose AIS is dark. Self-supervised learned dark-vessel re-ID across this modality set is, to the applicant's knowledge, absent from the literature — and it dissolves the scarce-label barrier that blocks supervised maritime fusion. It is novel, patentable foreground IP the applicant will protect — well beyond the prior patents. (a.2) Uncertainty-propagating, calibrated fusion. Per-modality evidential encoders feed a transformer association head over the set of concurrent observations; the fused identity score is conformally calibrated for distribution-free coverage — uncertainty is a first-class, propagated output, not a post-hoc number. This is the "advanced deep learning for spatiotemporal alignment, uncertainty propagation, and confidence scoring" the Challenge asks for. (a.3) Adaptive, constraint-aware perception. An agentic controller rewrites the edge perception graph (a text-defined MediaPipe .pbtxt graph) to swap networks under SWaP / threat conditions — a concrete realization of the Challenge's "learned, adaptive fusion across modalities rather than static aggregation." (b) Enhanced capability vs SOTA. Learned vs aggregated: a trained cross-modal association model replaces rule-based / threshold fusion — the exact distinction the Challenge draws ("learned, adaptive" vs "static aggregation"). Explainability: attention weights over modalities plus a per-output causation chain make every identity decision inspectable in one application, accreditable as a system property, beyond post-hoc XAI. Human-in-the-loop: analyst adjudication is a first-class signal that closes an active-learning loop — the applicant's shipped collaborative-intelligence pattern from the Orville, now applied to ISR decision-making and speed. Methodological pedigree: semi-supervised deep learning at archive scale on unlabeled acoustic streams (ORCA-SLANG; Sattar 2011, 95 % on ONC data) is the applicant's direct, peer-reviewed evidence the learning approach transfers to scarce-label maritime fusion. (c) Future potential. The "intermittent strong identifier self-supervises the rest" pattern generalizes to any multi-domain setting with a sometimes-present anchor — Arctic ISR (sat + RF + telemetry), Airborne Multi-Sensor (radar + EO/IR + telemetry), land / edge tactical (audio + video + sensor) — without redesign. The calibrated fusion head, the provenance-tracked entity graph, and the human-in-the-loop layer carry forward to Component 1b's second domain and to allied-interoperable maritime domain awareness.

PRC-3 - Impact of Proposed Solution

Articulate how your proposed solution meets the Impact criterion.

PRC-3 - Impact of Proposed Solution

(a) Addresses a stated capability gap. The Challenge names the gap: siloed multi-domain streams, rule-based aggregation, opaque outputs. The AIS-off ("dark vessel") problem in the Maritime Task Group Operations example is operationally costly — illegal fishing, sanctions evasion, ship-to-ship transfers, and Arctic sovereignty incursions all turn on AIS being disabled. Today this is handled with stovepiped per-modality pipelines and post-hoc explanation. Planetar replaces them with one learned cross-modal model that resolves identity from the modalities that persist after the beacon goes dark, and surfaces calibrated, explainable outputs. By restoring track continuity on adversaries that deliberately evade monitoring it reduces a surveillance vulnerability; by sharing the decision between AI correlation and analyst adjudication it increases speed of decision-making — the Challenge's two stated benefits. The capability is built on the applicant's own open-source code and new, Canadian-owned IP, and is open-source-replicable on commodity Linux — a sovereign capability interoperable with allies, aligned to the DND/CAF AI Strategy and Force Capability Plan priorities for ISR and C2. (b) Enhances S&T capability. Self-supervised cross-modal fusion in defence ISR: using an intermittent strong identifier (AIS) to self-supervise learned association across SAR/EO/acoustic/RF/text is a capability not present in current maritime ISR. Calibrated, explainable identity: propagated uncertainty plus an attention-grounded causal chain make outputs accreditable, addressing a known barrier to operational AI adoption. Methodological transfer: the applicant's archive-scale semi-supervised acoustic ML extends from cetacean-call to vessel-signature discrimination. Productionized entity resolution: retyped from research-entity to vessel-entity context, with full provenance on every resolved identity. (c) Matures the field. A generalizable pattern: "an intermittent strong identifier self-supervises the rest" transfers to Arctic, airborne, and land / edge multi-domain settings, and the applicant intends to publish it as a contribution to CAF multi-domain ISR. Lower barrier to accreditable AI: calibrated, lineage-bearing outputs make explainability a system property rather than an afterthought, supporting DND/CAF AI Strategy and ATO pathways. Durable output: applicant-controlled code and new, Canadian-owned IP outlast the contract window as papers and reference implementations.

PRC-4 - Feasibility and Approach of Proposed Solution

Articulate how your proposed solution meets the Feasibility and Approach criterion.

PRC-4 - Feasibility and Approach of Proposed Solution

(a) Achievable in practice. The substrate that makes the model demonstrable is already built and open-source (~13 k LOC of Planetar services): a provenance-tracked bus, a typed envelope, an entity-graph service (planetar-ontology, 30 tests; successor to the ~18-month doibio), an analyst shell, and four broker-integrated per-modality services (AIS, Sentinel-1 SAR, EO, hydrophone) that become the model's encoders. The 1a builds the new science on top: the self-supervised cross-modal fusion model, plus text and RF encoders. Public-data coverage: AIS (MarineCadastre / GFW) supplies both signal and the self-supervision labels; SAR (Sentinel-1, xView3); EO (Singapore Maritime / MODS); acoustic (ONC, ShipsEar, DeepShip); text (notices-to-mariners, port-state-control, OSINT — public); RF (best-effort public / synthetic — see risks). No GFP, no classified data. Compute fits ~\$18 K cloud — fine-tuning public encoders, not frontier-scale training. Solo execution is realistic: the applicant authored the substrate and has shipped comparable systems and peer-reviewed ML. (b) Well-reasoned approach. Two milestone stages, each with measurable exit criteria. Milestone 1: harden the substrate and the four ingresses and scaffold text + RF; build per-modality encoders into a shared embedding and the AIS-on co-occurrence training set; train the self-supervised cross-modal association model (the primary R&D). Milestone 2: propagate evidential uncertainty with conformal coverage and evaluate AIS-off re-identification; retype the vessel-entity graph; build the explainable, human-in-the-loop analyst surface (attention and causal-chain views, analyst adjudication); deliver the end-to-end Salish-Sea demonstration deployed live at planetar.ca for the evaluator to operate, with public-benchmark evaluation and the 1b package. The model fine-tunes public baselines, not frontier-scale training. (c) Risks + mitigations. RF data scarce publicly (medium-high): the RF encoder is built and evaluated on best-effort public / synthetic data; if unavailable, fall back to the five-modality model (AIS / SAR / EO / acoustic / text) — still well past the two-type bar, no overclaim. Self-supervision signal noisy (medium): supervised fine-tune on xView3 as an anchor; conformal coverage holds even if the score is weak. Acoustic vessel-transit GT sparse (medium): semi-supervised per ORCA-SLANG, discrimination via ShipsEar / DeepShip. Founder capacity (medium): a built-in schedule buffer is scope-cut insurance — cut RF or the waveform view before extending the timeline.

PRC-5 - Gender-based Analysis Plus (GBA Plus) of Proposed Solution

Articulate how your proposed solution meets the Gender-based Analysis Plus criterion.

Note: [Additional information on Gender-based Analysis Plus \(GBA Plus\).](#)

PRC-5 - Gender-based Analysis Plus (GBA Plus) of Proposed Solution

GBA Plus is applied to Planetar's technical solution along three axes: (i) operator-side accessibility and multilingual usability, (ii) training-data and fusion-model-output bias auditing, and (iii) deployment-side equity considerations. A cross-cutting principle: a human analyst stays accountable in the loop — the fusion model proposes, the operator adjudicates — guarding against automation bias that can fall unevenly on under-represented groups (small Indigenous-fleet craft, low-traffic regions). (i) Operator-side accessibility and multilingual usability — planned Milestone 2 deliverables. CAF operators span genders, ages, abilities, neurodivergence, and linguistic backgrounds. The shipped analyst shell (planetar-ui, React 19) will be enhanced in Milestone 2 against a documented accessibility checklist: keyboard navigation, ARIA roles on viewer panels and bus-message renderers, screen-reader compatibility, and a colour-vision-deficient-safe palette (confidence double-encoded with shape, not colour alone); the waveform viewer offers visual and audio playback so low-vision and audio-first analysts have equivalent access. Bilingual English / French via i18n in Milestone 2. The accessibility checklist is itself a 1a deliverable. (ii) Training-data and fusion-model-output bias auditing. Every dataset used for training or evaluation will be audited for population-representation bias along axes relevant to the maritime domain. Specifically: AIS coverage is denser in commercial shipping lanes than in Indigenous coastal fishing waters, which can encode a "default" of large-vessel patterns and degrade detection on small Indigenous-fleet craft; SAR datasets (xView3, SAR-Ship, FUSAR) are weighted toward Northern Hemisphere traffic; surface EO datasets (Singapore Maritime, MODS, SeaShips) over-represent equatorial shipping; hydrophone archives skew toward cetacean-research deployments off particular coastlines. The 1a deliverables include a dataset bias-audit report quantifying representation gaps and an evaluation protocol that stratifies model performance by vessel class, operator type, and geographic region — a measurable artefact, not a generality. (iii) Deployment-side equity considerations. The architecture's SWaP profile (commodity-Linux laptop-class hardware, no kernel bypass, no FPGA) means the system can deploy in low-bandwidth or remote settings — including Indigenous coastal communities, small-port operations, and Arctic coast-guard auxiliary stations — at the same fidelity as it deploys in central operations centres. Equity is a structural property of the architecture, not a deployment afterthought. Implementation in the 1a: the dataset bias-audit report, the operator-accessibility checklist, and multilingual-readiness verification are explicit Milestone 2 work-plan deliverables — GBA Plus applied to the technical solution, not deferred to a follow-on phase.

PRC-6 - Alignment of Desired Outcomes

Articulate how your proposed solution meets the Alignment of Desired Outcomes criterion.

Note: If no Desired Outcomes are identified in the Challenge statement, any information provided by the Offeror in this section will not be evaluated, and Offerors will receive 15 points.

PRC-6 - Alignment of Desired Outcomes

Planetar covers all five of the Challenge's desired outcomes through one learned cross-modal fusion model and the substrate it runs on: (1) Spatiotemporal alignment, uncertainty propagation, confidence scoring across modalities. The model encodes each modality and aligns observations in a shared spatiotemporal embedding — alignment is learned, not hand-keyed. Per-modality evidential heads propagate uncertainty through the fusion, and the fused identity score is calibrated by conformal prediction for distribution-free coverage. Deep-learning alignment, propagated uncertainty, and calibrated confidence — exactly as specified. Covered. (2) Entity resolution + dynamic knowledge graph for persistent cross-domain tracking. Resolved observations populate a dynamic vessel-entity graph that is a substantial development beyond the applicant's prior named-inventor patents (US 10,936,582, US 11,442,952; Salesforce-assigned) and the applicant's ~18-month implementation: canonical Vessel many identifications (MMSI, IMO, RF / acoustic signature, hull-OCR, report mentions) provenance records. Covered. (3) Policy-aware fusion with provenance tracking, full lineage across classification levels. Every observation and every inference carries source and causation_id; outputs are traceable to inputs bit-for-bit, replayable from any historical cursor. Classification-level metadata travels with each item and consumers filter by policy at access time (Protected B at 1a; a multi-level design path). Lineage is structural, not bolted on. Covered. (4) Scalable real-time pipelines with explainable outputs for operator trust. The model runs on a provenance-tracked substrate on commodity hardware; explainability is native — attention weights over modalities plus click-through from any output back to its raw inputs — and a human analyst is a first-class participant whose adjudication calibrates trust and feeds an active-learning loop. Covered. (5) SWaP and compute limits incorporated for edge deployment. The substrate is a small-footprint C binary + zero-copy envelope + browser shell, running on laptop-class commodity Linux — no FPGA, no kernel-bypass NIC, no specialised silicon; and the model's perception front-end is a runtime-rewritable MediaPipe graph that adapts networks to power / threat limits — constraint-aware by construction, addressing the Edge Fusion for Tactical Units example. Covered.

PRC-7 - Alignment of Proposed Solution Costs

In the following sections, you will be asked to provide your detailed work plan and financial proposal. The breakdown of the Financial Proposal must align with the work proposed, and specifically:

- a. Is realistic for the technical approach proposed;
- b. Is appropriate with, and is reasonably relative to:
 - i. each labour category proposed per milestone;
 - ii. the number of labour hours proposed per milestone;
 - iii. the types and categories of materials, equipment, and other direct costs;
 - iv. each task for which travel and costs incurred is proposed, the travel costs incurred are explained and reasonable.

Solution Progression to Follow-on Components

Zax Analytics intends to progress this solution into Component 1b (Develop, TRL 4-5) and beyond. Component 1a delivers a demonstrated proof-of-concept at TRL 3: a trained, calibrated, self-supervised cross-modal fusion model that re-identifies dark vessels — vessels that disable their AIS transponders to evade tracking — by fusing AIS, synthetic-aperture radar, electro-optical imagery, passive acoustics, radio-frequency

emissions and textual maritime reports. It is validated on public data and operable live at planetar.ca, where a Technical Authority can drive it during evaluation. S&T work under Component 1b advances the capability from proof-of-concept to operationally-relevant maturity along four lines. 1) Productionize and harden the model. We move from public and synthetic data to operationally representative streams; harden ingest, monitoring and graceful degradation when sensors drop; and add robustness to adversarial behaviour such as AIS spoofing and identity swapping. The human-in-the-loop adjudication captured in 1a becomes an online active-learning loop, so operator corrections continuously refine the model — a Collaborative Intelligence design in which the analyst is empowered by the AI and the AI improves from the analyst. 2) Operational validation with a CAF partner. We evaluate dark-vessel re-identification against operator-defined scenarios and metrics on representative data in a controlled environment (TRL 4), then in an operationally relevant environment (TRL 5). Uncertainty remains conformally calibrated and every output remains traceable to its inputs, so results are accreditable rather than opaque. 3) Extend to a second multi-domain setting. The core idea — using an intermittent strong identifier (AIS, here) to self-supervise the other modalities — generalizes without redesign. The same pattern applies to Joint ISR for Arctic operations (satellite imagery, RF and telemetry) and to airborne multi-sensor platforms (radar, EO/IR and telemetry for detecting stealth or spoofed assets), broadening from the maritime beachhead to multi-domain situational awareness. 4) Edge deployment and accreditation. We adapt the perception front-end for size-, weight- and power-constrained tactical-edge hardware (the Edge Fusion for Tactical Units use case) and degraded-connectivity operation; add multi-level classification handling so the system operates across security levels with full lineage; and begin a formal accreditation / Authority-to-Operate readiness package. End-state of Component 1b: a validated, operationally-relevant cross-modal fusion capability at TRL 5 — component and system validated in a relevant environment — delivering calibrated, explainable, policy-aware multi-domain situational awareness across maritime and at least one additional domain, with an accreditation-readiness package and a clear path to TRL 6+ operational deployment. Capabilities and benefits for Canada. The capability restores track continuity on vessels that deliberately evade monitoring, directly supporting maritime domain awareness against illegal fishing, sanctions evasion, ship-to-ship transfers and Arctic sovereignty incursions. It increases the speed of decision-making by sharing the load between AI correlation and human adjudication, and reduces system vulnerability by replacing brittle, siloed, single-modality pipelines with one auditable, provenance-tracked fusion model. It aligns with the DND/CAF AI Strategy and Force Capability Plan priorities for ISR, C2 and operational resilience. Because it is grounded in the applicant's own open-source code and new, Canadian-owned IP, and is open-source-replicable on commodity hardware, it is a sovereign Canadian capability that interoperates with allies (NATO and Five-Eyes maritime domain awareness). Advancement to market and commercialization. The capability is dual-use. Beyond the CAF, the same cross-modal fusion and entity-resolution engine serves civil maritime domain awareness for the Canadian Coast Guard, Fisheries and Oceans Canada, Transport Canada and port authorities, and allied defence-and-security customers for multi-domain ISR. Zax Analytics will mature the 1a and 1b deliverables into a deployable product, with a data flywheel in which each confirmed re-identification becomes a labelled example that improves the model — a compounding advantage that incumbents tied to batch, rule-based fusion struggle to

replicate. The progression path is Component 1a (Conceive, TRL 3), to Component 1b (Develop, TRL 4-5), to operational deployment and procurement (TRL 6+), positioning Canada with a sovereign, exportable multi-domain fusion capability.

Work Plan and Deliverables

This section will be used to assess the criteria and supporting information of PRC-4.

Each row of the following tables must contain specific activities and deliverables (work). Work must be distributed into two (2) milestone stages. Work for each milestone stage must render the project to a point of progress sufficient to enable the Technical Authority (TA) to assess the potential of the Project Description through completion of each Milestone thereafter.

In the tables that follow, Offerors must specify the work activities to be performed. Decompose the work activities to a level that exposes all risk factors, that allows accurate estimation of the requirements, and the schedule duration for each task activity (**maximum 800 characters per table entry**).

Milestone tables

Funds are limited to a maximum allowance of **CAN\$250,000** (excluding applicable taxes) per contract for a performance period of up to six (6) months.

The cost proposal for Milestone 1 must be no greater than **70%** of the total cost for the proposed solution. A table must be filled in for each Milestone.

Financial Proposal

This section will be used to assess the criteria and supporting information for SC-1 and PRC-7.

In the tables that follow, Offerors must specify the various costs proposed for each milestone (excluding applicable taxes).

The two (2) milestone tables must be completed. Cost(s) indicated in the "Other Costs" table must fall within the definition of Eligible Costs in Section 3.7 of the Solicitation document.

The total cost proposal must not exceed the individual contract maximum of **CAN\$250,000** (excluding applicable taxes).

The Total Firm Milestone 1 Price must be no greater than **70%** of the total cost proposal (Milestones 1 & 2 combined). The Financial Proposal must align with the work proposed.

Note: All costs identified in the proposal must be in Canadian dollars.

Milestone 1

Milestone 1 requires the following:

Progress reporting (Milestone 1) to be completed and delivered in accordance with the Statement of Work (SOW).

Total Performance Period (in weeks) 13 (weeks)				
Work Activities & Risks and Mitigation				
Activity	Description of Deliverable	Estimated Level of Effort (Weeks)	Risk(s): Description, probability and impact	Risk mitigation strategy(ies)
Train the self-supervised cross-modal fusion model — a transformer association head operating over the set of concurrent observation embeddings — to learn vessel identity from the co-occurrence signal.	A trained cross-modal fusion model that associates concurrent observations to a common vessel identity, validated on held-out public data.	4.00	The model may under-perform on modalities with scarce labels (medium probability). Impact: lower association accuracy for those modalities, though calibration in Milestone 2 bounds the resulting uncertainty.	Build on the fine-tuned public baselines from Activity 3 rather than training at frontier scale; the applicant's peer-reviewed semi-supervised deep learning on hydrophone archives is direct precedent that the approach transfers to the scarce-label regime.
Harden the real-time data substrate — the message bus, the typed provenance-carrying envelope, and write-ahead-log replay — and place it under continuous integration with regression tests so replay is bit-for-bit reproducible. This is the transport layer every modality publishes onto, fixed before any	A hardened bus + typed-envelope + WAL-replay substrate under continuous integration, with regression gates passing and replay reproducible bit-for-bit. The shared transport every ingest service publishes onto.	1.00	The substrate is already built, so risk is low; hardening could expose latent replay edge cases (low probability). Impact: minor rework contained within this activity, no effect on the downstream schedule.	Lock the substrate behind regression tests before the ingest and model work depends on it, so any replay defect surfaces at the transport layer rather than downstream in training.

ingest or model work depends on it.				
Harden the four existing per-modality ingest services (AIS, Sentinel-1 SAR, electro-optical, hydrophone) and add two new ingest paths — text (notices-to-mariners typed, port-state-control reports, open-source maritime intelligence) and radio-frequency emissions — so every input arrives as a typed, time-stamped, provenance-tagged observation on public data, under continuous integration.	All six modalities (AIS, SAR, EO, acoustic, RF, text) streaming typed, provenance-tagged observations from public sources onto the bus, with regression gates passing. This is the input layer the fusion model consumes.	2.00	RF is the hardest modality to source publicly (medium-high probability). Impact: the model fuses five modalities instead of six, slightly narrowing coverage — but it still exceeds the required two heterogeneous types, so impact on the core objective is low.	Source RF from best-effort public and synthetic data. If public RF proves insufficient, proceed with the five-modality model (AIS, SAR, EO, acoustic, text) — no overclaim, still well past the essential two-type bar. Scope the RF ingest path so it can be dropped without affecting the other five.
Build per-modality encoders that project each observation — SAR chip, EO chip, acoustic spectrogram, AIS kinematics, RF signature, and report text — into one shared embedding space, fine-tuning established public baselines	A set of per-modality encoders producing comparable embeddings in one shared space — the representation layer the fusion model operates over.	2.00	Modalities with scarce public labels (RF, text) may yield weak embeddings (medium probability). Impact: lower-quality embeddings for those modalities, partly recovered by the	Fine-tune established public baselines rather than training from scratch; the applicant's peer-reviewed semi-supervised deep learning on hydrophone archives is precedent that the approach

(CLIP, I-JEPA, triplet-loss re-identification families) rather than training at frontier scale.			co-occurrence training in Activity 5.	transfers to scarce-label modalities.
Apply the trained fusion model to re-identify vessels whose AIS has gone dark, producing cross-modal re-identification candidates, each carrying its per-modality supporting evidence; demonstrate end-to-end on held-out public data.	Dark-vessel re-identification candidates with per-modality supporting evidence, demonstrated end-to-end on held-out public data — the Milestone 1 capability proof.	2.00	Re-identification precision on dark vessels may be modest before calibration (medium probability). Impact: candidate confidence is not yet bounded — addressed directly by the conformal calibration in Milestone 2.	Treat Milestone 1 output as ranked candidates with evidence, not hard decisions; Milestone 2 adds distribution-free confidence bounds before any operational claim.
Construct the self-supervision training set from AIS-on periods: when a vessel is broadcasting AIS, its identity automatically labels the concurrent observations from the other modalities in the same place and time, producing cross-modal training pairs at no labelling cost. Anchor with the publicly labelled xView3 SAR/AIS dark-vessel dataset and	A labelled cross-modal co-occurrence dataset built from public AIS-on periods, with measured co-occurrence-pairing precision — the training data for the fusion model.	2.00	The self-supervision signal is noisy where AIS coverage is sparse or intermittent (medium probability). Impact: weaker training pairs could reduce the accuracy of the learned cross-modal association.	Anchor the training set with the publicly labelled xView3 SAR/AIS dark-vessel dataset, and quantify label quality (precision of the co-occurrence pairing) before training so noisy pairs are filtered or down-weighted.

quantify label quality before training.				
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Milestone 1 - Labour			
Category	Labour (h)	Rate (\$/h)	Total (\$)
Proposer Scientist (PhD CS/ML)	440	140.00	61600.00
Total Labour Cost (\$)			61600.00

Milestone 1 - Materials			
Description	Quantity	Unit Cost (\$)	Total (\$)
MacBook Pro	1	7924.00	7924.00
Total Materials Cost (\$)			7924.00

Milestone 1 - Other Costs		
Other cost	Description	Cost (\$)
Cloud/GPU	Cloud GPU/CPU compute for Sentinel-1 tile fetching, training and fine-tuning the fusion model and per-modality encoders, evaluation runs, and hosting the live planetar.ca demonstrator.	9000.00
Software / tools	Development and operations tooling: IDE, observability/monitoring, continuous-integration runner minutes, and ancillary paid developer services.	1000.00
Datasets / licences	Reserve for paid dataset access if a public dataset proves insufficient; core training and evaluation data are public and free.	500.00
Total Labour Cost (\$)		10500.00

TOTAL FIRM MILESTONE 1 PRICE (\$)	
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(including labour, materials, other costs, travel and cost incurred, excluding applicable taxes)

80024.00

Milestone 2

Milestone 2 requires the following:

Progress reporting (Milestone 2) to be completed and delivered in accordance with the Statement of Work (SOW).

Total Performance Period (in weeks) 13 (weeks)				
Work Activities & Risks and Mitigation				
Activity	Description of Deliverable	Estimated Level of Effort (Weeks)	Risk(s): Description, probability and impact	Risk mitigation strategy(ies)
Propagate per-modality uncertainty through the fusion model: per-modality evidential heads are carried through the transformer association head so that every fused re-identification score arrives with a principled uncertainty estimate rather than as a bare point score. This makes model confidence a first-class, propagated output, ready for calibration.	A fusion model that emits, for each re-identification, a fused score together with a propagated per-modality uncertainty estimate.	2.00	Evidential heads may be poorly calibrated on scarce-label modalities such as RF and text (medium probability). Impact: the raw uncertainty estimates could be over- or under-confident before calibration.	Feed the propagated uncertainty into the conformal calibration step (Activity 2), which guarantees coverage regardless of the raw estimate — a distribution-free back-stop, so confidence is sound after calibration even if the raw heads are imperfect.
Calibrate the fused re-identification score with conformal				

prediction, converting it into a distribution-free confidence bound so each re-identification carries a coverage-guaranteed confidence interval instead of an uncalibrated number. Validate the empirical coverage on held-out public data.	A calibrated fusion model whose re-identification outputs carry distribution-free confidence intervals at a stated, empirically validated coverage level.	1.00	Calibration could be weak if the underlying re-identification scores are noisy (low probability). Impact: confidence bounds would be wider than ideal, but they remain valid.	Conformal prediction guarantees the stated coverage even when the base score is weak — a distribution-free back-stop, so calibrated confidence is assured regardless of score quality.
Evaluate AIS-off (dark-vessel) re-identification on public benchmarks, producing an evaluation report stratified by vessel class and geographic region so performance is measured rather than asserted, with calibrated confidence reported on each result.	A public-benchmark evaluation report on dark-vessel re-identification, stratified by vessel class and geographic region, with calibrated confidence on each result.	1.00	Public dark-vessel ground truth is uneven across vessel classes and regions (medium probability). Impact: thin strata yield wider confidence intervals and weaker per-class conclusions.	Anchor on the publicly labelled xView3 SAR/AIS dark-vessel dataset and report per-stratum support, so thin strata are flagged rather than over-claimed.
Retype the vessel-entity graph for the maritime domain so each resolved identity carries full provenance — which observation, from which sensor, at what	A vessel-domain entity graph in which each canonical		Retyping the entity model to the vessel domain may surface schema mismatches with the existing	Reuse the ~18-month entity-resolution implementation as the starting point and retype incrementally, validating each

time, supported the match — making every re-identification traceable to its raw inputs. This builds on the applicant's existing ~18-month entity-resolution implementation.	identity links to its supporting observations with full per-match provenance.	1.00	implementation (low probability). Impact: minor rework of the type mappings, contained within this activity.	mapping against the bus envelope schemas before moving on.
Build the explainable, human-in-the-loop analyst surface on the existing shell: attention-weighted evidence views showing which modalities and observations drove each re-identification, and click-through from any output back to its raw inputs, so an operator can adjudicate each candidate. Capture the operator's accept/reject/correct decisions as labelled feedback for future model refinement.	An operator console driving the live model, with attention-based explanations, click-through to raw evidence, and adjudication capture that feeds an active-learning loop.	4.00	Viewer scope could overrun the schedule (medium probability). Impact: less time remaining for the final integrated demonstration.	Ship the map, timeline and entity-card views first; the raw-evidence (waveform/chip) view ships as a minimal renderer. Non-essential viewer scope is cut before the schedule slips.
Integrate the full pipeline into an end-to-end demonstration: a synthetic Salish Sea scenario in which a vessel disables AIS	A live demonstrator at			A built-in schedule buffer;

during a Sentinel-1 satellite pass and is re-identified across SAR, EO and acoustic, replayable bit-for-bit. Deploy a live system at planetar.ca that the Technical Authority can operate during evaluation. Run a public-benchmark evaluation and prepare the Component 1b proposal.	planetar.ca the evaluator can operate, a replayable end-to-end Salish Sea dark-vessel scenario, a public-benchmark evaluation report, and the Component 1b proposal package.	4.00	Solo-founder capacity is the main schedule risk (medium probability). Impact: feature scope at risk near the deadline.	non-essential scope (the RF modality, the raw-evidence viewer) is cut before the timeline extends. The substrate is already built, so effort concentrates on the model and demo, not plumbing.
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Milestone 2 - Labour

Category	Labour (h)	Rate (\$/h)	Total (\$)
Proposer Scientist (PhD CS/ML)	485	140.00	67900.00
Total Labour Cost (\$)			67900.00

Milestone 2 - Other Costs

Other cost	Description	Cost (\$)
Cloud GPU/Compute	Cloud GPU/CPU compute for model training, calibration and evaluation runs, the public-benchmark evaluation, and hosting the live planetar.ca demonstrator the evaluator can operate.	9000.00
Software / tools	Development and operations tooling: observability/monitoring, continuous-integration runner minutes, and	500.00

	ancillary paid developer services. AI agentic coding solution.	
Datasets / Licences	Reserve for paid dataset access if a public dataset proves insufficient; core training and evaluation data are public and free.	500.00
Total Labour Cost (\$)		10000.00

TOTAL FIRM MILESTONE 2 PRICE (\$) (including labour, materials, other costs, travel and cost incurred, excluding applicable taxes)	77900.00
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Location of Work

The Offeror will perform the work, including travel and living, in the following location(s)
(Maximum 256 characters per table entry)

Note: If none, Canada will leave this article blank or delete it.

City	Province / Territory	Country
Victoria	British Columbia	Canada

Language of Work

The contract will require work to be performed in either one or both of the [Official Languages of Canada](#). The Offeror is to indicate its capability:

~ *English*

Glossary

The following terms are used in this Proposal and have the following definitions:

If acronyms and/or uncommon terms are utilized, the Offeror is to add and define them below.

Note: If none, Canada will leave this article blank or delete it.

Acronyms and/or uncommon terms	Definitions
Provenance (lineage)	the recorded chain of which inputs produced a given output, enabling audit and explanation.
Embedding	a learned numeric vector representation in which related observations lie close together.

Self-supervised learning	training a model from structure in unlabelled data; here, AIS-on co-occurrence supplies free labels.
C2	Command and Control
ISR	Intelligence, Surveillance and Reconnaissance.
ONC	Ocean Networks Canada, the University of Victoria-operated cabled ocean observatory, including hydrophones.
RF	Radio Frequency, radio-frequency emissions used for detection and signals intelligence.
EO	Electro-Optical, visible-spectrum (camera) imagery.
Sentinel-1	the European Space Agency's free, public C-band SAR satellite mission.
SAR	Synthetic-Aperture Radar, satellite or airborne radar imaging that operates day or night and through cloud.
AIS	Automatic Identification System, a transponder system that broadcasts a vessel's identity, position, course and speed.
CLIP	Contrastive Language-Image Pre-training, a public method for learning a shared cross-modal embedding.
xView3	a public dark-vessel-detection dataset and competition (Sentinel-1 SAR co-located with AIS).
Cross-modal fusion	combining data from different sensor types into a single model or representation.
I-JEPA	Image-based Joint-Embedding Predictive Architecture, a public self-supervised learning method.
IMO Number	International Maritime Organization number, a permanent unique identifier assigned to a vessel's hull.
Transformer	a neural-network architecture that uses attention over a set of inputs.

MMSI	Maritime Mobile Service Identity, the unique numeric identifier transmitted by a vessel's AIS transponder.
WAL	Write-Ahead-Log : an append-only log recording every event so the system's state can be replayed exactly.
Conformal prediction	a statistical method giving distribution-free confidence bounds on a model's outputs.

Reference Documents

If the work details include references to documents critical for understanding the basis of the work, the Offeror is to fully describe these documents here.

Note: If none, Canada will leave this article blank or delete it.

Title	Author	Publication Date	Relevance to the Work
ORCA-SLANG: An Automatic Multi-Stage Semi-Supervised Deep Learning Framework for Large-Scale Killer Whale Call Type Identification	C. Bergler, M. Schmitt, A. Maier, H. Symonds, P. Spong, S. R. Ness, G. Tzanetakis, E. Noeth	2021-08-30	Applicant's co-authored semi-supervised deep learning at archive scale on unlabelled hydrophone streams — evidence the proposed self-supervised approach transfers to the scarce-label maritime regime.
The Orchive: A System for Semi-Automatic Annotation and Analysis of a Large Collection of Bioacoustic Recordings	Steven Ness	2014-01-01	The applicant's doctoral work — archive-scale semi-supervised machine learning on a 20,000-hour, 30-year OrcaLab hydrophone archive.

Automatic Event Detection for Long-Term Monitoring of Hydrophone Data	F. Sattar, P. Driessen, G. Tzanetakis, S. R. Ness, W. Page	2011-08-24	Applicant's co-authored, peer-reviewed event detection on Ocean Networks Canada hydrophone data at 95% accuracy — direct precedent for the acoustic modality and for machine learning on continuous, scarcely-labelled streams.
US Patent 11,442,952 B2 — User interface for commerce architecture	Steven R. Ness, one of 11 named inventors (assignee: Salesforce, Inc. — named-inventor credit only, not the assignee)	2022-09-14	Canonical-data-model mapping with matching / reconciliation rules for unified profiles — identity-resolution prior art the applicant contributed to; secondary background credential, distinct from and superseded by the project's new fusion-model IP.
US Patent 10,936,582 B2 — Integrated entity view across distributed systems	Steven R. Ness, one of 19 named inventors (assignee: salesforce.com, inc. — named-inventor credit only, not the assignee)	2021-03-03	The integrated-entity-view / entity-resolution architecture the project takes as a starting point and develops substantially beyond — the maritime entity graph that resolves cross-modal observations to canonical vessels with full provenance.